The death of Latin: Detecting the demographic drivers of European vernacularization 1450-1900

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The global spread of languages has been determined by a complex interplay of migration, contact, and language shift (Mufwene 2025). A particular type of language spread is found in European Early Modern vernacularization (Tahko & Marjanen 2024). The educated elite moved away from Latin dominance, and local communities were empowered to communicate in regional varieties (Figure 1). The newly formed communities have a direct link to the formation of nation states and form a backbone of European politics up to now (Anderson 1983; Burke 2004). Here, we study the formation of local written-language communities through large bibliographic databases that capture a near-comprehensive record (ca. 10 million books) of works published across Europe in 1450-1900. Specifically, we try to understand the role of individuals and demographic processes in these transformations: How much of the change is driven by language shifts among the members of these communities, how much is due to new people being able to join these communities?

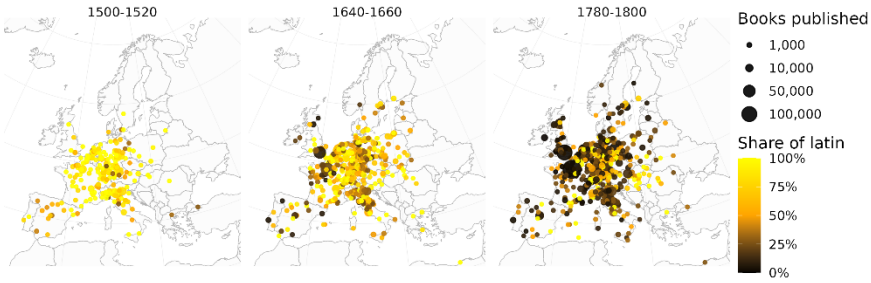

Figure 1: The share of Latin among books published in Europe (n = 10,442,056). Each point denotes a city, scaled to the number of books published in the 20-year period. Data: HPB

We rely on two massive data sources, the Heritage of Printed Books (HPB) and collected national bibliographies (Estonian, Finnish, Czech). These datasets build on the effort of generations of librarians and bibliophiles to gather and systematize all known published books and have come recently to be used for cultural analytic workflows (Lahti et al. 2019). This data allows us to reconstruct multilingual publishing communities in high detail. For each author in a community, we can see in which languages they were published throughout their career and the linguistic choices made by the publishers in different cities.

We analyze the large-scale dynamics of language use in books using the Price equation, adopted from evolutionary ecology (Sobchuk & Beheim 2025; Beheim & Baldini 2012). The Price equation allows us to decompose trends of language use into underlying forces. Here, we decompose each temporal trend into three effects: entrances of individuals, their exits, and the individual change of language preferences (i.e., switching a language mid-career).

Figure 2 shows an exemplary comparison between two transformations, (A) from Latin to German in the 18th century in the borders of modern Germany, and (B) from German to Estonian in the 19th century in the borders of modern Estonia. For the expansion of German, individual change dominates, as in a situation of language shift due to new administrative prestige. For the expansion of Estonian, the emergence of new writers dominates, as the barriers to join the written-language communities are reduced by political changes and educational reforms.

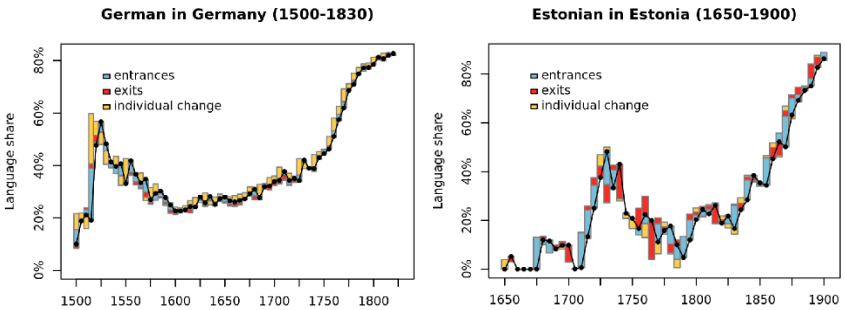

Figure 2: Two example decomposition plots. 1) The use of German vernacular in modern-day borders of Germany. There are two major patterns of growth: first, a quick leap during the Lutheran reformation due equally to entrances of new contributors and individual change (1500–1530), and second, a gradual growth mostly due to individual change (1730–-1780). 2) The use of Estonian in the modern-day borders of Estonia. We see two major periods of growth: temporarily in 1720–1755, and more persistently in 1830-1900. Both changes are mostly explained by population turnover: that is, by individuals entering and exiting the community. Data: HPB, Estonian National Bibliography

The ecology inspired approach to bibliographic data here follows the attempts to combine language use data with explicit models of demographic processes (Baxter & Croft 2016; Blythe & Croft 2021). Decomposition analysis shows the

story of vernacularization to be driven by both demographic processes and language shift. The differences between the regions provide a key also to a better understanding of the origins of modern European language diversity.

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